

Jordan Thompson

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EDUCATION

BASc in Biomedical Engineering | University of British Columbia

May 2026

TECHNICAL SKILLS

Hardware R&D & Test Equipment: Wearable sensors (IMU, sEMG, PPG), PCB design, hands-on lab equipment setup, battery and firmware troubleshooting

Programming & Data Review: Python, C++, MATLAB, Git, simple data review summaries (dashboards, spreadsheets), Google Workspace tools, iOS/Android platforms

-Research Operations: Human subjects research protocols, data quality checks, SOP development, ISO 13485

RELEVANT EXPERIENCE

Research Assistant | HuMBL, UBC

Sep 2025–May 2026

- Built a MATLAB contraction-detection algorithm using the Teager-Kaiser Energy Operator (TKEO) across all 64 HD-sEMG channels; automated a process that took 3 months deviating less than 1% from manual ground truth.
- Developed a gradient descent optimizer to determine sensor shift between sessions by minimizing RMSE across spatially mapped signal features (inter-channel correlation, mean envelope, mean frequency), enabling robust cross-session alignment for clinical stroke rehabilitation data.

Biomedical Test Engineering Co-op | Kardium Inc.

May–Aug 2025

- Developed and executed test methods for the Globe® Mapping & Ablation System, evaluating cleaning and sterilization efficacy via TOC and FT-IR spectroscopy to demonstrate ISO 10993 biocompatibility compliance for regulatory submission.
- Analyzed biological safety datasets using non-parametric statistical methods in Minitab; identified root causes of production anomalies and collaborated cross-functionally to drive process corrections.

Undergraduate Research Assistant | Bioaugmentative Interfaces Lab, UBC

May 2024–Apr 2025

- Developed a C++ closed-loop control system for an SMA-actuated implantable bladder device; benchmarked multiple control algorithms to minimize power draw and limit external surface temperature, using thermocouple and infrared camera measurements to pinpoint components requiring additional insulation.
- Conducted pre-clinical validation in rodent surgery and cadaver models, iterating device designs based on in situ bladder voiding performance.

Mechatronics Co-op Student | Bioaugmentative Interfaces Lab, UBC

May–Sep 2024

- Designed and iterated multiple CAD mold geometries (3D-printed in PLA) to cast SMA actuators into silicone elastomer structures; selected elastomer material for its dual role as thermal insulator and soft biocompatible implant interface.
- Fabricated a custom test fixture (artificial rat bladder, tubing, stand) and programmed a C++ Arduino flow meter to quantify flow velocity and actuator output force, enabling objective comparison of actuator designs.
- Designed a custom PCB (ublox BMD-340) and programmed BLE firmware in Zephyr OS, including custom devicetree, GATT service profile, and GAP configuration to achieve low-standby-power wireless control via smartphone.

Engineering & Maintenance Co-op | New Beta Innovation

May–Oct 2023

- Maintained calibration for 50+ instruments with zero workflow disruption; authored SOPs for equipment sterilization and supported new equipment validation under an ISO 13485 QMS.

SELECTED PROJECTS

Pediatric Injection Pain Relief Device | Academic Project

Sep 2025–Present

- Designed a custom PCB motor driver and PLA housing (3D-printed, modular) for an open-source vibro-cooling device targeting children with JIA (~42 injections/year); verification testing confirmed 162 Hz peak vibration (within 20–200 Hz clinical target) and stable surface cooling of 4.7–7.4°C over 10 minutes, both derived from clinician/patient stakeholder interviews and gate control theory.

Wearable Fall Detection System | Academic Project

Jan–Apr 2025

- Built a Python ML pipeline for multi-sensor IMU data: 0.1 Hz high-pass + 150/250 Hz low-pass filters, segmented into 10-second rolling windows (5-second overlap) to capture full 6-second fall events; tuned a Random Forest classifier via GridSearch with 5-fold cross-validation, achieving 63.3% PPV with a conservative any-sensor-triggers-alarm decision rule to maximize patient safety.